

1. General Description

The EM74CBTLV3253 provides a dual 1-of-4 high-speed multiplexer/demultiplexer with two common select inputs (S0, S1) and two output enable inputs ($\overline{1OE}$, $\overline{2OE}$). The low ON resistance of the switch allows inputs to be connected to outputs without adding propagation delay or generating additional ground bounce noise. When pin $\overline{nOE}$ = LOW, one of the four switches is selected (low-impedance ON-state) with pins S0 and S1. When pin $\overline{nOE}$ = HIGH, all switches are in the high-impedance OFF-state, independent of pins S0 and S1.

Schmitt trigger action at control input makes the circuit tolerant to slower input rise and fall times across the entire V_{CC} range from 2.3 V to 3.6 V.

To ensure the high-impedance OFF-state during power-up or power-down, $\overline{nOE}$ should be tied to the V_{CC} through a pull-up resistor. The minimum value of the resistor is determined by the current-sinking capability of the driver.

This device is fully specified for partial power-down applications using I_{OFF} . The I_{OFF} circuitry disables the output, preventing the damaging backflow current through the device when it is powered down.

2. Features and Benefits

- Supply voltage range from 2.3 V to 3.6 V
- 2.4 Ω switch connection between two ports
- CMOS low power dissipation
- I_{OFF} circuitry provides partial Power-down mode operation
- Latch-up performance exceeds 250 mA
- Rail to rail switching on data I/O ports
- Complies with JEDEC standard:
 - JESD8-5 (2.3 V to 2.7 V)
 - JESD8B/JESD36 (2.7 V to 3.6 V)
- ESD protection:
 - HBM ANSI/ESDA/JEDEC JS-001 Class 3A exceeds 6000 V
 - CDM ANSI/ESDA/JEDEC JS-002 Class C3 exceeds 2000 V
- Multiple package options

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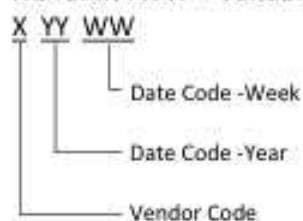
3. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EM74CBTLV3253D	CBTLV3253 YYYWW	SOP-16L	plastic small outline package; 16 leads; body width 3.9 mm	3000
EM74CBTLV3253PW	CBTLV3253 YYYWW	TSSOP-16L	plastic thin shrink small outline package; 16 leads; body width 4.4 mm	3000

MARKING INFORMATION

NOTE: XYYWW = Vendor Code and Date Code.



4. Function Diagram

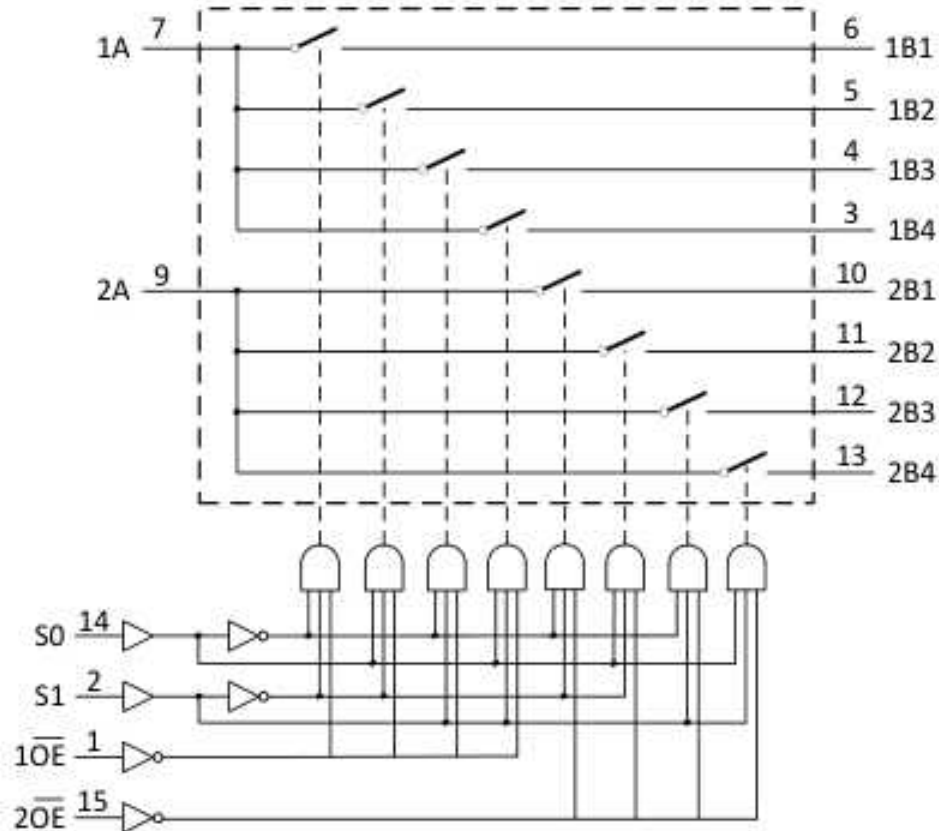
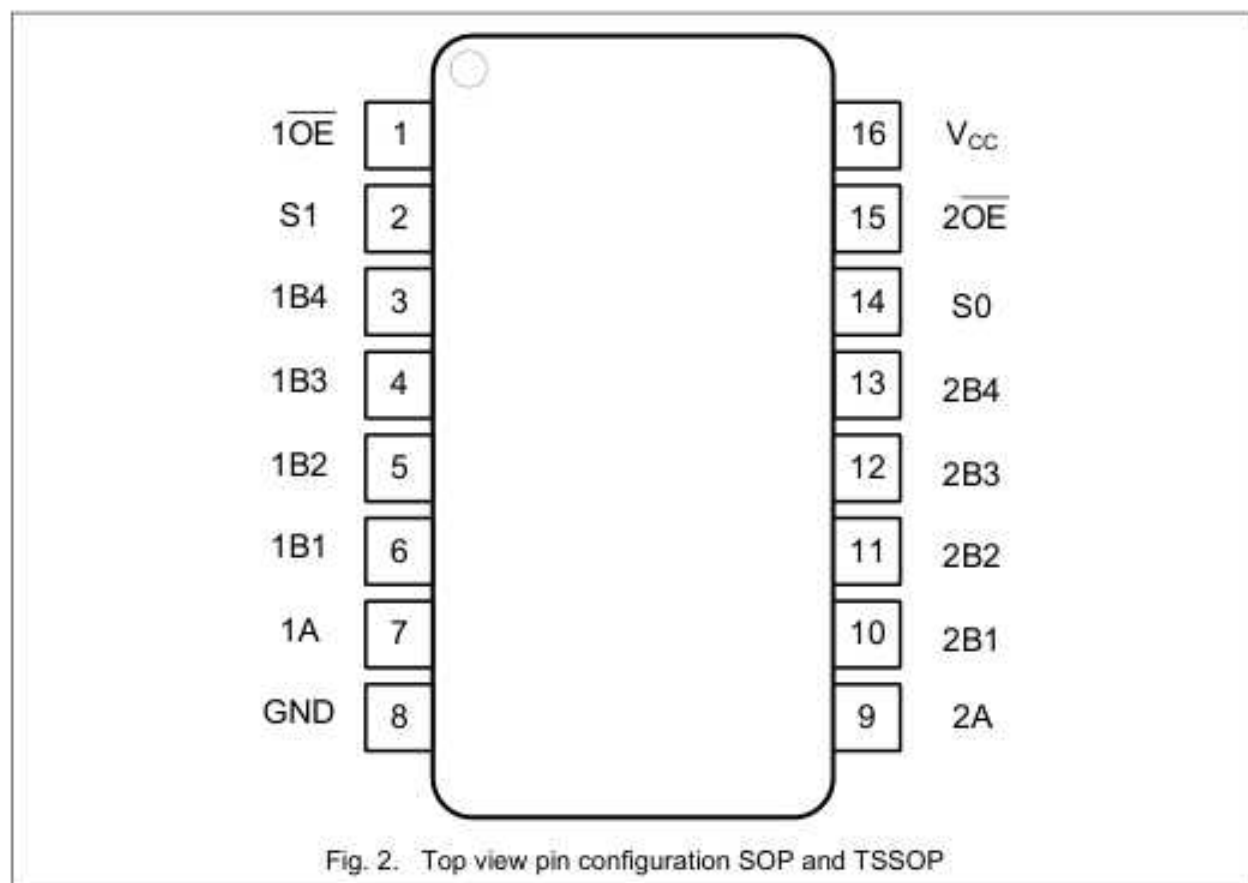


Fig. 1. Logic diagram

5. Pinning Information

5.1. Pinning



5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
1OE, 2OE	1, 15	Output enable input (active LOW)
S0, S1	14, 2	Select input
1B1 to 1B4	6, 5, 4, 3	B input/output
2B1 to 2B4	10, 11, 12, 13	B input/output
GND	8	Ground (0 V)
1A, 2A	7, 9	A input/output
VCC	16	Supply voltage

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; X = don't care.

Inputs				Function switch
1OE	2OE	S1	S0	
X	H	X	X	Disconnect 2A and 2Bn
H	X	X	X	Disconnect 1A and 1Bn
L	L	L	L	1A to 1B1 and 2A to 2B1
L	L	L	H	1A to 1B2 and 2A to 2B2
L	L	H	L	1A to 1B3 and 2A to 2B3
L	L	H	H	1A to 1B4 and 2A to 2B4

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	4.6	V
I_{IK}	input clamping current	$V_I < -0.5$ V	-50		mA
V_I	input voltage	control inputs [1]	-0.5	4.6	V
V_{SW}	switch voltage	enable and disable mode [2]	-0.5	$V_{CC}+0.5$	V
I_{SK}	switch clamping current	$V_I < -0.5$ V	-50		mA
I_{SW}	switch current	$V_{SW} = 0$ V to V_{CC}		± 128	mA
I_{CC}	supply current			100	mA
I_{GND}	ground current		-100		mA
P_{tot}	total power dissipation	$T_{amb} = -40$ °C to $+125$ °C		500	mW
T_{sig}	storage temperature		-65	150	°C

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

[2] The switch voltage ratings may be exceeded if switch clamping current ratings are observed.

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		2.3	3.6	V
V_I	input voltage		0	3.6	V
V_{SW}	switch voltage	enable and disable mode	0	V_{CC}	V
T_{amb}	ambient temperature		-40	125	°C
$\Delta t/\Delta V$	input transition rise and fall rate	$V_{CC} = 2.3 \text{ V to } 3.6 \text{ V}$ [1]	0	200	ns/V

[1] Applies to control signal levels.

9. Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
V_{IH}	HIGH-level input voltage	$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$	1.7			1.7		V
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$	2.0			2.0		V
V_{IL}	LOW-level input voltage	$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$			0.7		0.7	V
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$			0.9		0.9	V
I_I	input leakage current	pin $\overline{nOE}$, S_n ; $V_{CC} = 3.6 \text{ V}$; $V_I = \text{GND to } V_{CC}$			± 1		± 20	μA
$I_{S(OFF)}$	OFF-state leakage current	$V_{CC} = 3.6 \text{ V}$; see Fig. 3			± 1		± 20	μA
$I_{S(ON)}$	ON-state leakage current	$V_{CC} = 3.6 \text{ V}$; see Fig. 4			± 1		± 20	μA
I_{OFF}	power-off leakage current	$V_{CC} = 0 \text{ V}$; $V_I \text{ or } V_O = 0 \text{ V to } 3.6 \text{ V}$			± 10		± 50	μA
I_{CC}	supply current	$V_{CC} = 3.6 \text{ V}$; $V_I = V_{CC} \text{ or GND}$; $V_{SW} = V_{CC} \text{ or GND}$; $I_O = 0 \text{ A}$;			10		50	μA
ΔI_{CC}	additional supply current	pin $\overline{nOE}$, S_n ; $V_{CC} = 3.6 \text{ V}$; [2] $V_I = V_{CC} - 0.6 \text{ V}$; $V_{SW} = \text{GND or } V_{CC}$			300		2000	μA
C_i	input capacitance	pin $\overline{nOE}$; $V_{CC} = 3.3 \text{ V}$; $V_I = 0 \text{ V to } 3.3 \text{ V}$		1.5				pF
$C_{S(OFF)}$	OFF-state capacitance	$V_{CC} = 3.3 \text{ V}$; $V_I = 0 \text{ V to } 3.3 \text{ V}$						
		A port		20				pF
		B port		5				pF
$C_{S(ON)}$	ON-state capacitance	$V_{CC} = 3.3 \text{ V}$; $V_I = 0 \text{ V to } 3.3 \text{ V}$		28				pF

[1] All typical values are measured at $T_{AMB} = 25 \text{ }^\circ\text{C}$.

[2] One input at 3 V, other inputs at V_{CC} or GND.

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Table 7. Resistance R_{ON}

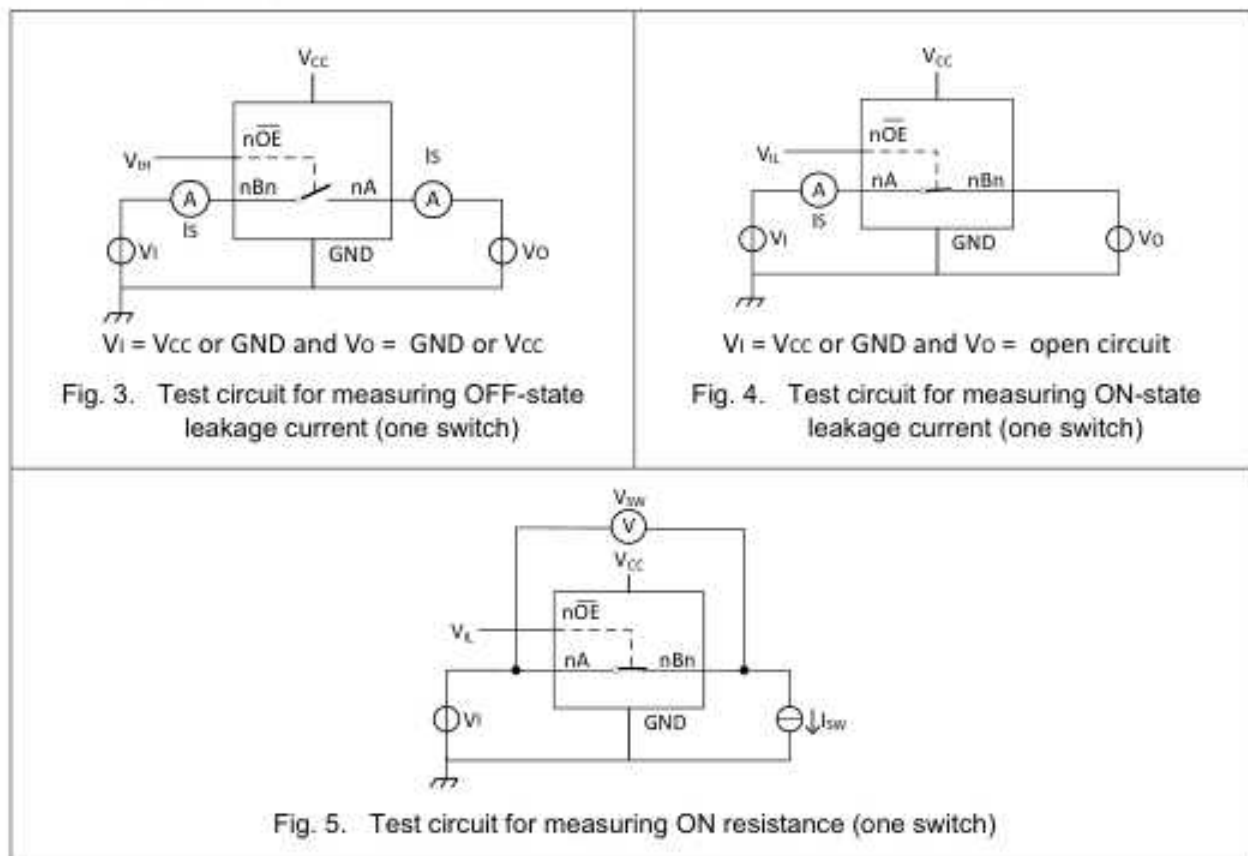
At recommended operating conditions; voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 5.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
R_{ON}	ON resistance	$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}; \quad [2]$						
		$I_{SW} = 64 \text{ mA}; V_I = 0 \text{ V}$		2.8	5		6	Ω
		$I_{SW} = 24 \text{ mA}; V_I = 0 \text{ V}$		2.8	5		6	Ω
		$I_{SW} = 15 \text{ mA}; V_I = 1.7 \text{ V}$		6.6	10		12	Ω
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V};$						
		$I_{SW} = 64 \text{ mA}; V_I = 0 \text{ V}$		2.5	4.5		5.5	Ω
		$I_{SW} = 24 \text{ mA}; V_I = 0 \text{ V}$		2.5	4.5		5.5	Ω
		$I_{SW} = 15 \text{ mA}; V_I = 2.4 \text{ V}$		4.4	8		10	Ω

[1] All typical values are measured at nominal V_{CC} and $T_{amb} = 25 \text{ °C}$.

[2] Measured by the voltage drop between the A and B terminals at the indicated current through the switch. ON-state resistance is determined by the lower of the voltages of the two (A or B) terminals.

9.1. Test circuit



10. Dynamic Characteristics

Table 8. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 8.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
t_{pd}	propagation delay	nA to nBn or nBn to nA; see Fig. 6 [2][3]						
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$			0.3		0.4	ns
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$			0.3		0.4	ns
t_{en}	enable time	$\overline{nOE}$ to nA or nBn; see Fig. 7[4]						
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$	1.0	2.9	4.4	1.0	5.3	ns
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$	1.0	2.6	4.2	1.0	5	ns
		Sn to nA or nBn; see Fig. 7 [4]						
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$	1.0	3	4.5	1.0	5.4	ns
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$	1.0	2.5	4	1.0	5	ns
t_{dis}	disable time	$\overline{nOE}$ to nA or nBn; see Fig. 7[5]						
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$	1.0	2.8	5	1.0	6	ns
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$	1.0	2.7	4.9	1.0	6	ns
		Sn to nA or nBn; see Fig. 7 [5]						
		$V_{CC} = 2.3 \text{ V to } 2.7 \text{ V}$	1.0	3.2	5.3	1.0	6.2	ns
		$V_{CC} = 3.0 \text{ V to } 3.6 \text{ V}$	1.0	3.1	5.2	1.0	6.1	ns
$f_{ -3dB }$	-3 dB frequency response	$V_{CC} = 3.3 \text{ V}; R_L = 50 \Omega$; [6]		240				MHz

[1] All typical values are measured at $T_{amb} = 25 \text{ °C}$ and at nominal V_{CC} .

[2] The propagation delay is the calculated RC time constant of the typical on-state resistance of the switch and the load capacitance, when driven by an ideal voltage source (zero output impedance).

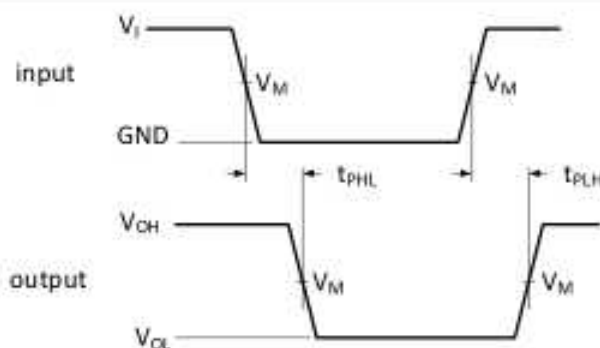
[3] t_{pd} is the same as t_{PLH} and t_{PHL} .

[4] t_{en} is the same as t_{ZHL} and t_{ZHL} .

[5] t_{dis} is the same as t_{HLZ} and t_{HLZ} .

[6] f_i is biased at $0.5 \times V_{CC}$.

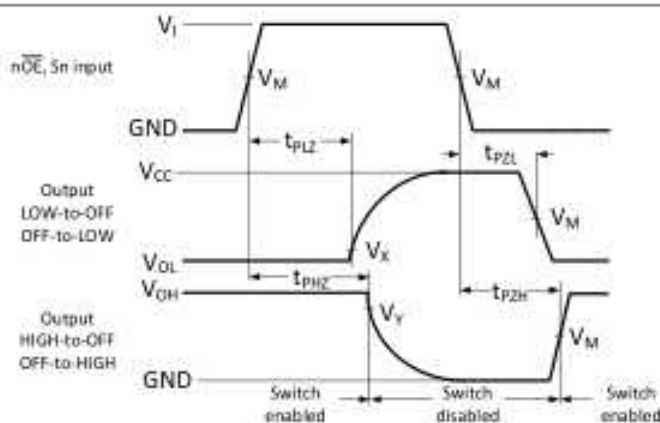
10.1. Waveforms and test circuit



Measurement points are given in Table 9.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 6. The input (nA or nBn) to output (nBn or nA) propagation delays



Measurement points are given in Table 9.

V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 7. Enable and disable times

Table 9. Measurement points

Supply voltage	Input	Output		
V_{CC}	V_M	V_M	V_X	V_Y
2.3 V to 2.7 V	$0.5V_{CC}$	$0.5V_{CC}$	$V_{OL} + 0.15V$	$V_{OH} - 0.15V$
3.0 V to 3.6 V	$0.5V_{CC}$	$0.5V_{CC}$	$V_{OL} + 0.3V$	$V_{OH} - 0.3V$

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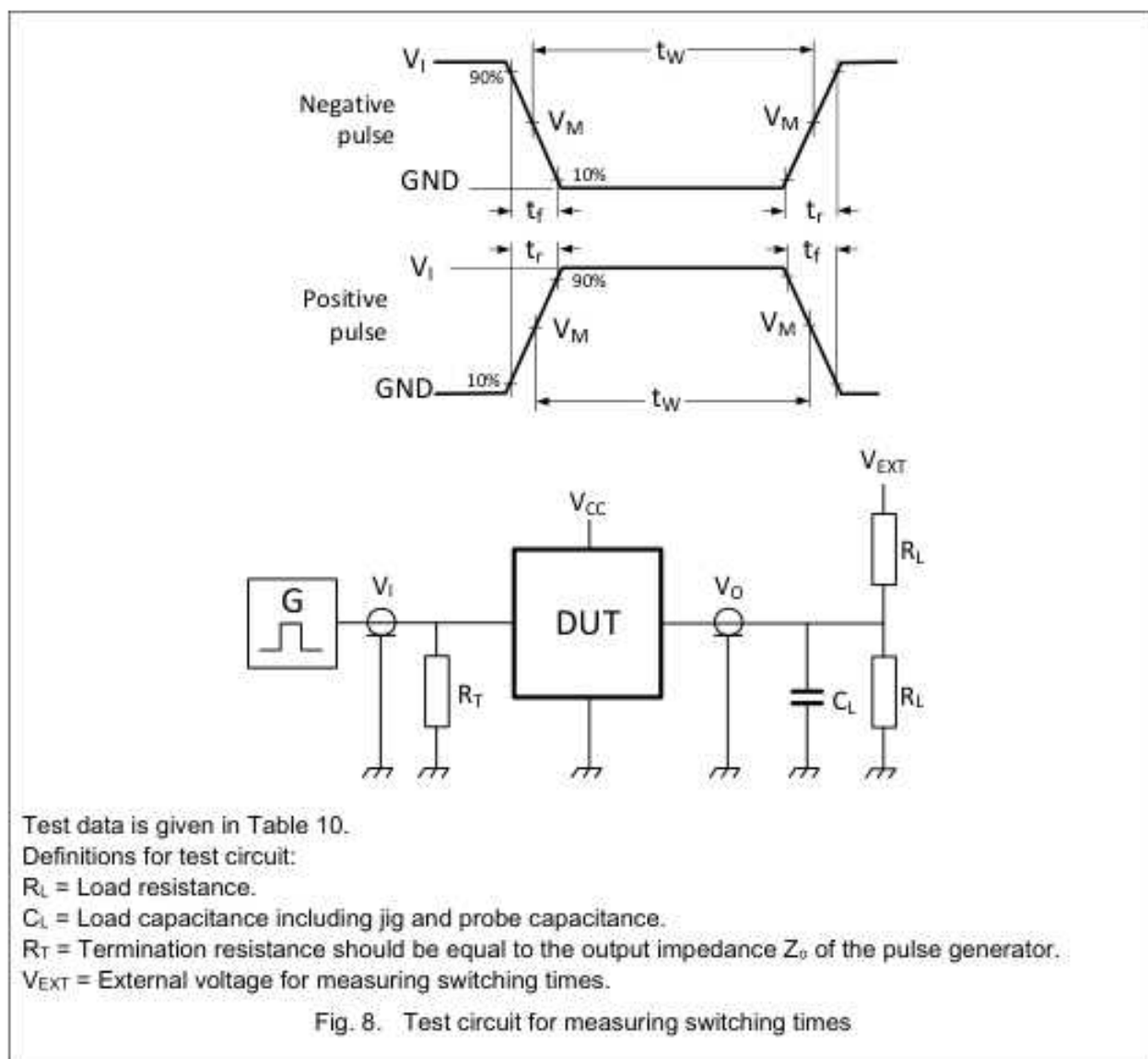
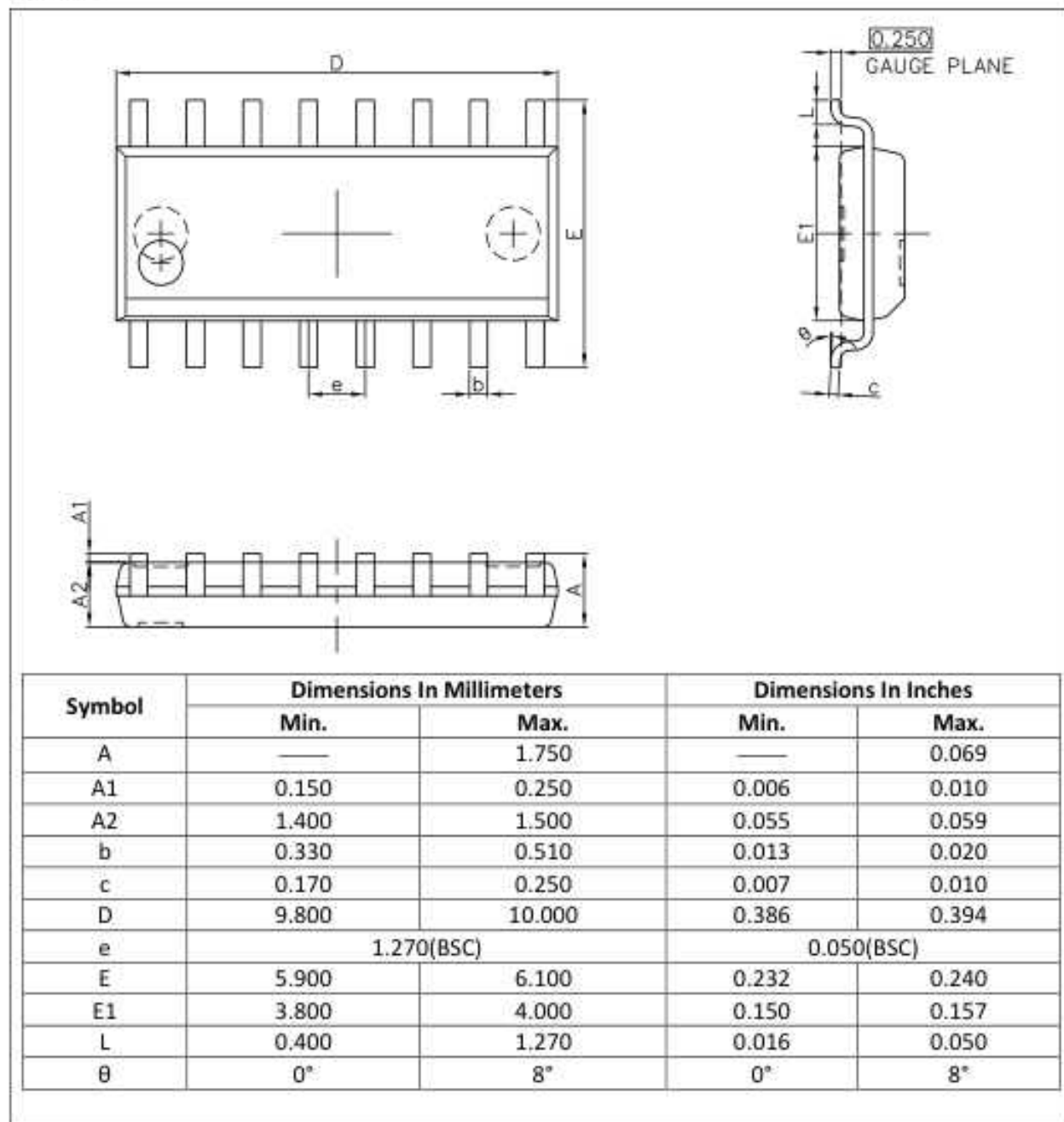


Table 10. Test data

Supply voltage	Input		Load		V_{EXT}		
V_{CC}	V_I	$t_r = t_f$	C_L	R_L	t_{PLH}, t_{PHL}	t_{PLZ}, t_{PHZ}	t_{PZL}, t_{PLZ}
2.3 V to 2.7 V	V_{CC}	≤ 2.0 ns	15 pF	500 Ω	open	GND	$2 \times V_{CC}$
3.0 V to 3.6 V	V_{CC}	≤ 2.0 ns	15 pF	500 Ω	open	GND	$2 \times V_{CC}$

11. Package Outline

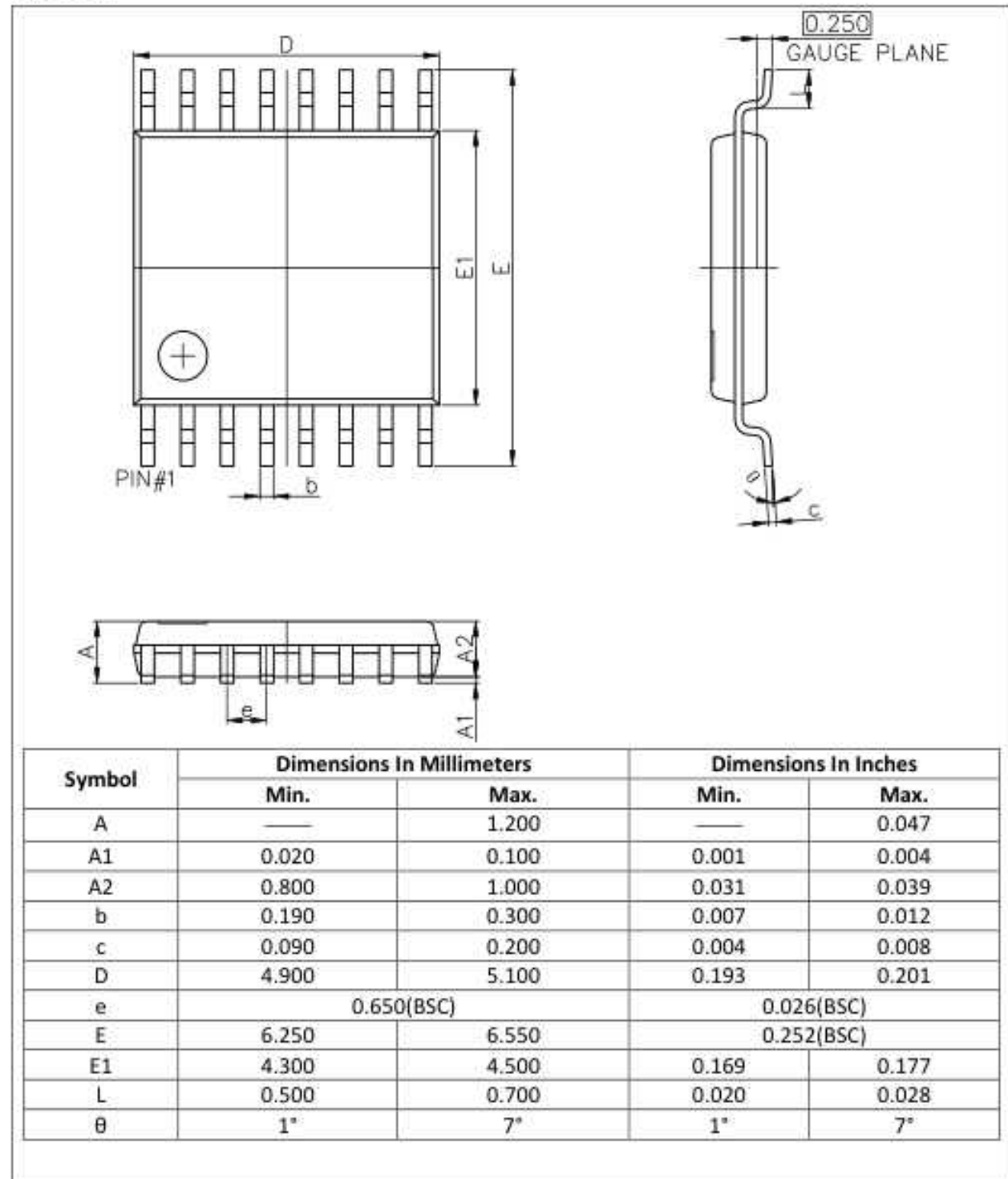
SOP-16L



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TSSOP-16L



12. Tape and Reel Information

12.1. Carrier tape dimensions

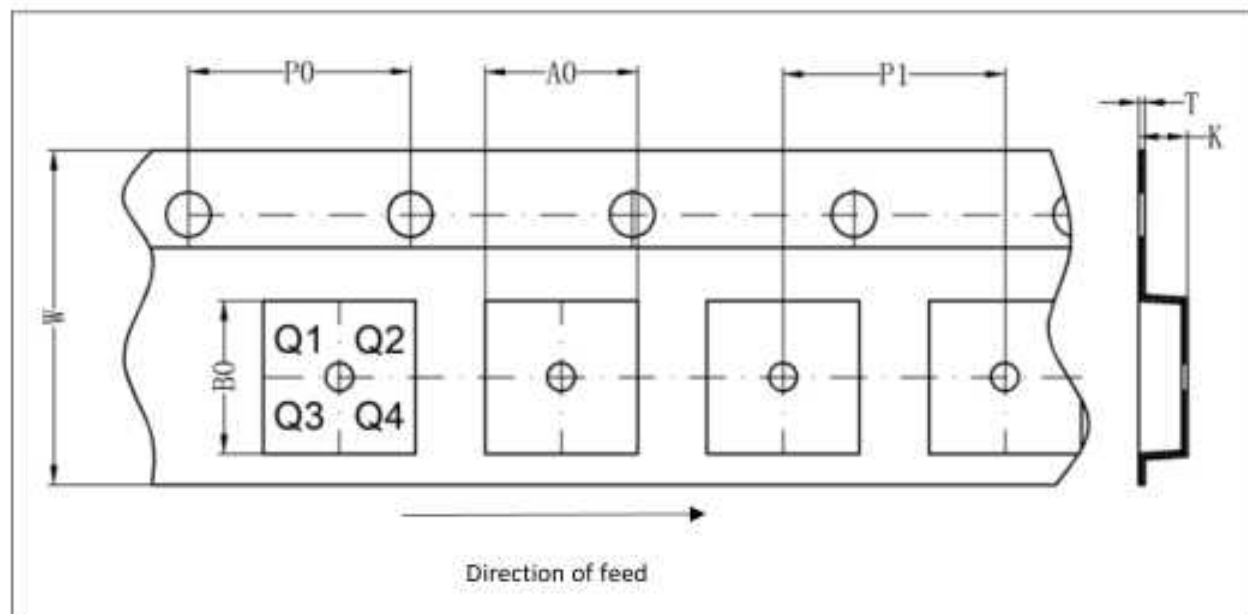


Table 11. Carrier tape dimensions

Package version	A0(mm)	B0(mm)	K0(mm)	T(mm)	P1(mm)	W(mm)	P0(mm)	PIN 1
SOP-16L	6.5	10.45	2.1	0.22	8	16	4	Q1
TSSOP-16L	6.7	5.45	1.6	0.25	8	12	4	Q1

12.2. Reel and box dimensions

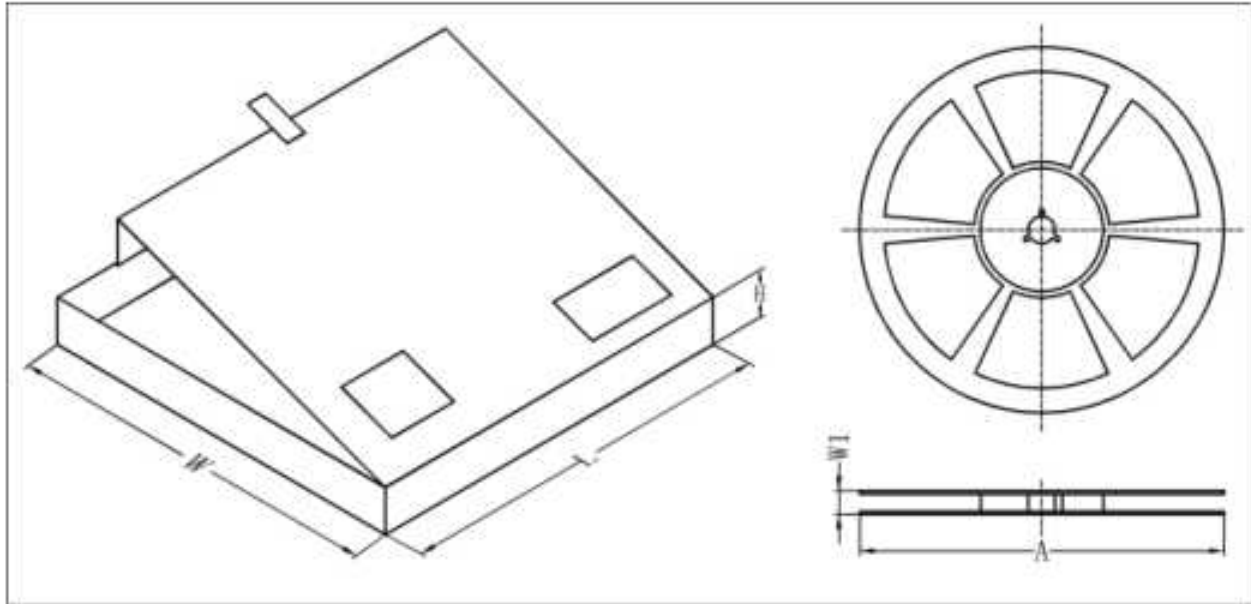


Table 12. Dimensions and quantities

Package version	Type NO. ending	Reel Dimension A (mm)	Reel Width W1 (mm)	MPQ (pcs)	Reels per box	Outer box dimensions L×W×H(mm)
SOP-16L	D	330	22.4	3000	1	358x340x50
TSSOP-16L	PW	330	18.4	3000	1	358x340x50

13. Abbreviations

Table 13. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model

14. Revision History

Table 14. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74CBTLV3253 Rev. 1.0	Jul 02, 2026	Product datasheet		